3.2 Parallel Lines and Transversals

Name _____ Period ____ Date _____

Geometry 3.2 (30-minute plan) · Student sheet

I can…

- use corresponding angles to find angle measures when lines are parallel.
- use alternate interior, alternate exterior, and consecutive interior angles to find measures.
- prove a theorem about parallel lines.

Vocabulary · fill in the missing word

Corresponding angles — angles in the same position at the two _____ of a transversal

Alternate interior angles — angles between the two lines on opposite sides of the _____

Alternate exterior angles — angles outside the two lines on opposite sides of the _____

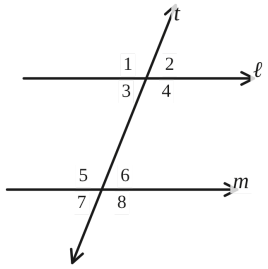
Consecutive interior angles — angles between the two lines on the same side of the _____

Part 1 · Notes and examples (fill in each note, then work the example)

Corresponding Angles Theorem: if two parallel lines are cut by a transversal, then corresponding angles are _____.

Example 1

$\ell \parallel m$ and $m\angle 1 = 115°$. Find $m\angle 5$ and $m\angle 8$.

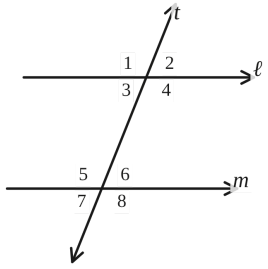


Alternate Interior and Alternate Exterior Angles Theorems: these pairs are also _____ when the lines are parallel.

Consecutive Interior Angles Theorem: if two parallel lines are cut by a transversal, then consecutive interior angles are _____.

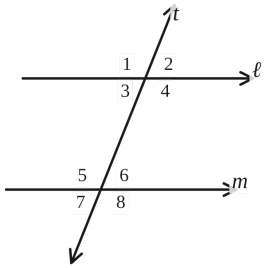
Example 2

$\ell \parallel m$, $m\angle 4 = (3x + 10)°$, and $m\angle 6 = (2x + 20)°$. Find x and both angles.



Example 3

$\ell \parallel m$, $m\angle 3 = (4x - 15)°$, and $m\angle 6 = (2x + 35)°$. Find x and the angle measure.



Part 2 · Partner practice

Partner practice

$\ell \parallel m$, $m\angle 1 = (6x - 8)°$, and $m\angle 8 = (4x + 22)°$. Find x and the angle measure.

